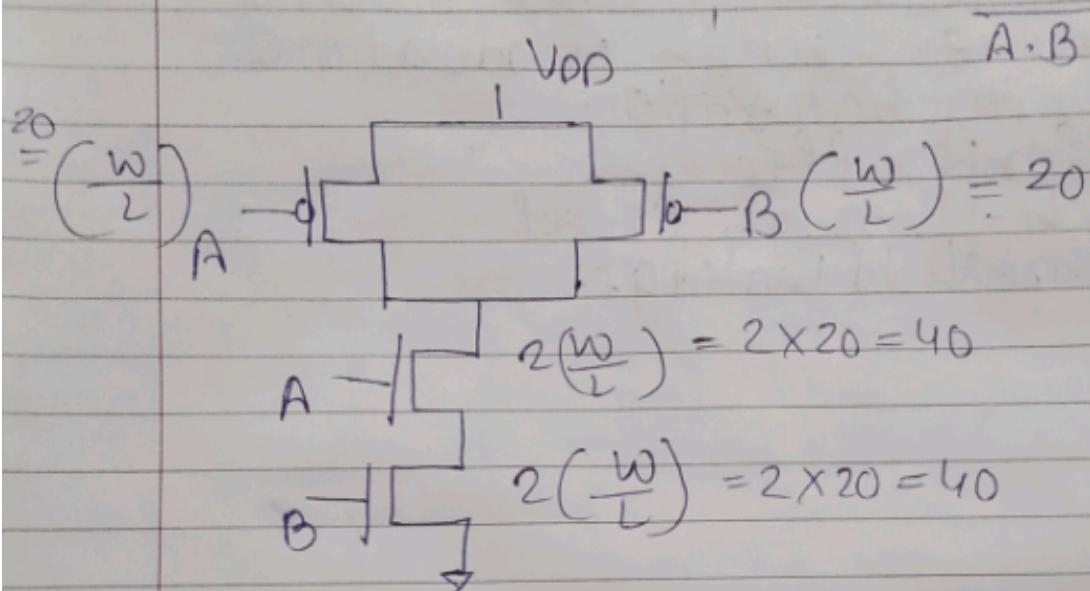
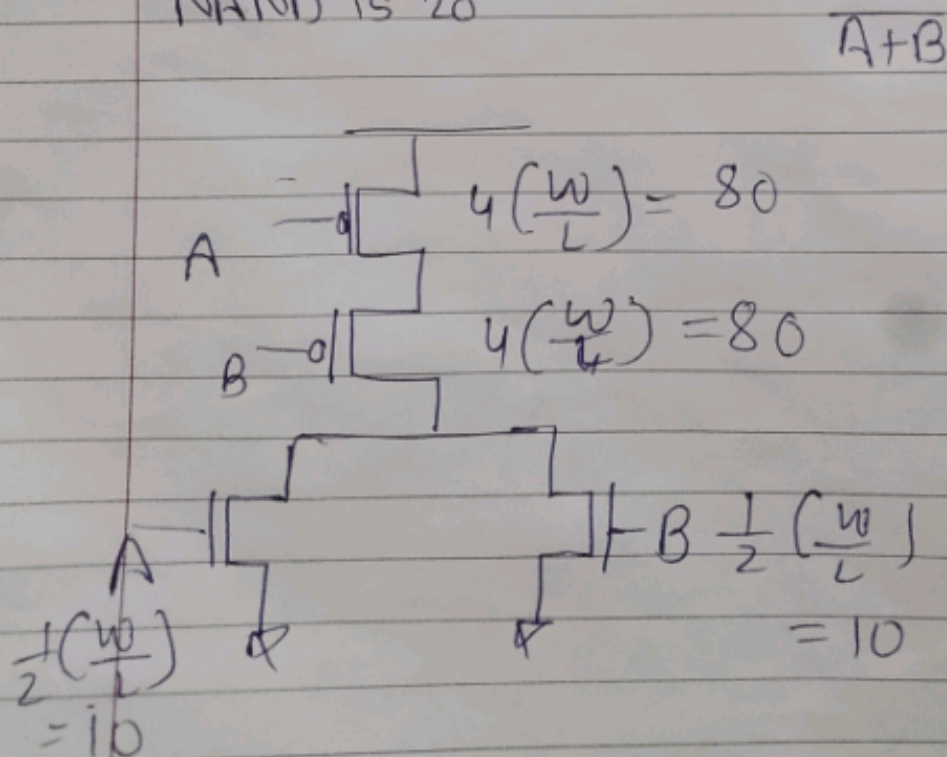


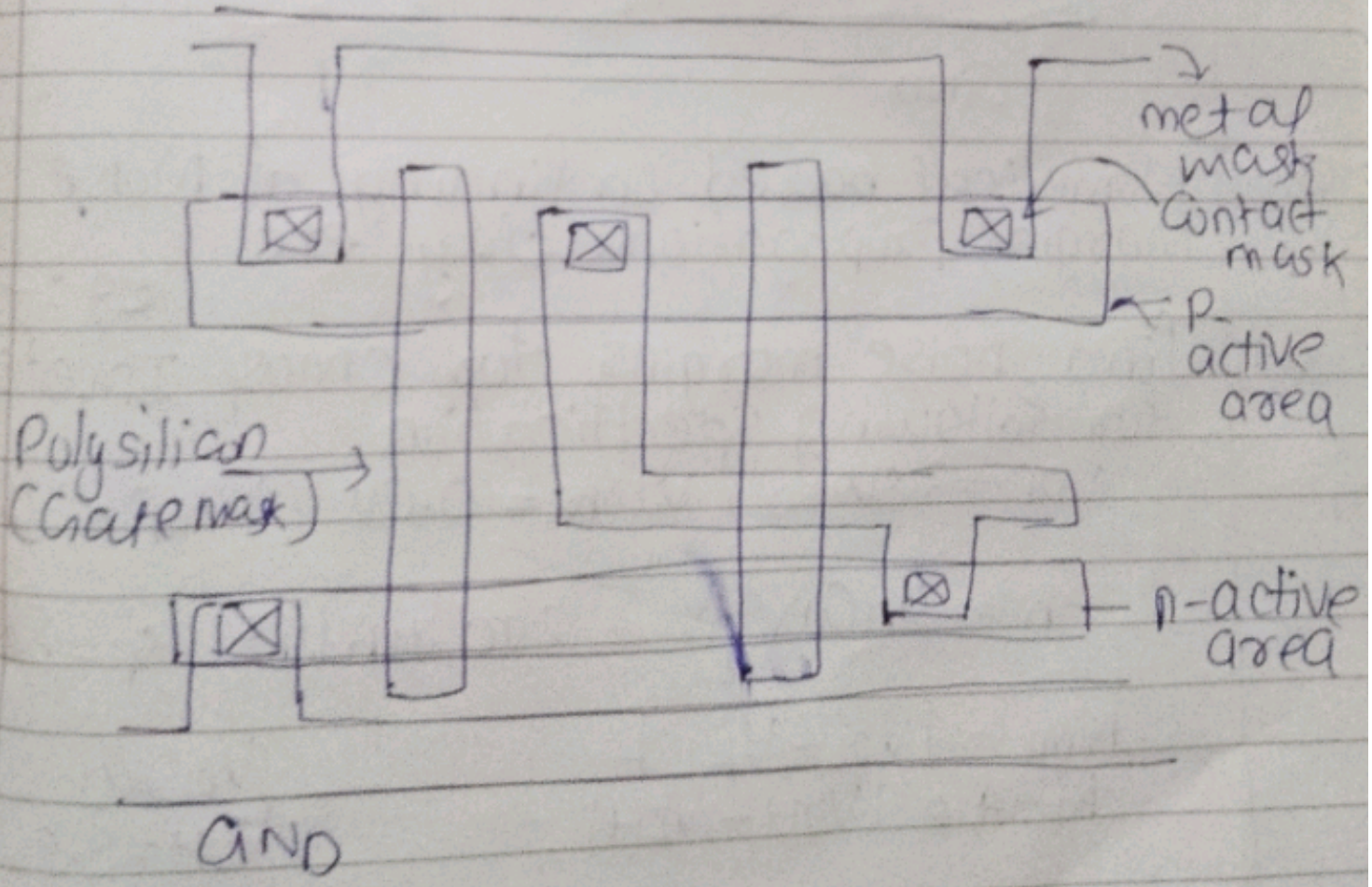
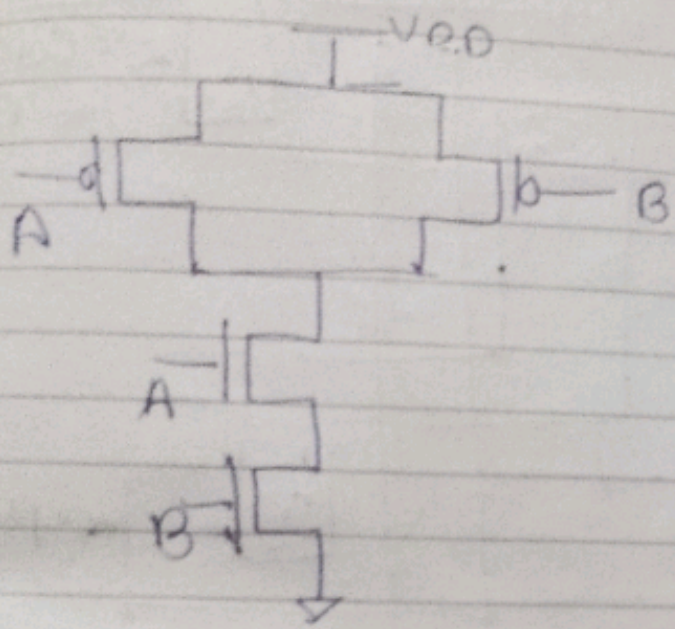
30. Draw two input NAND gate and its equivalent inverter if $(\frac{w}{L})$ ratio of all transistors in NAND gate is 20



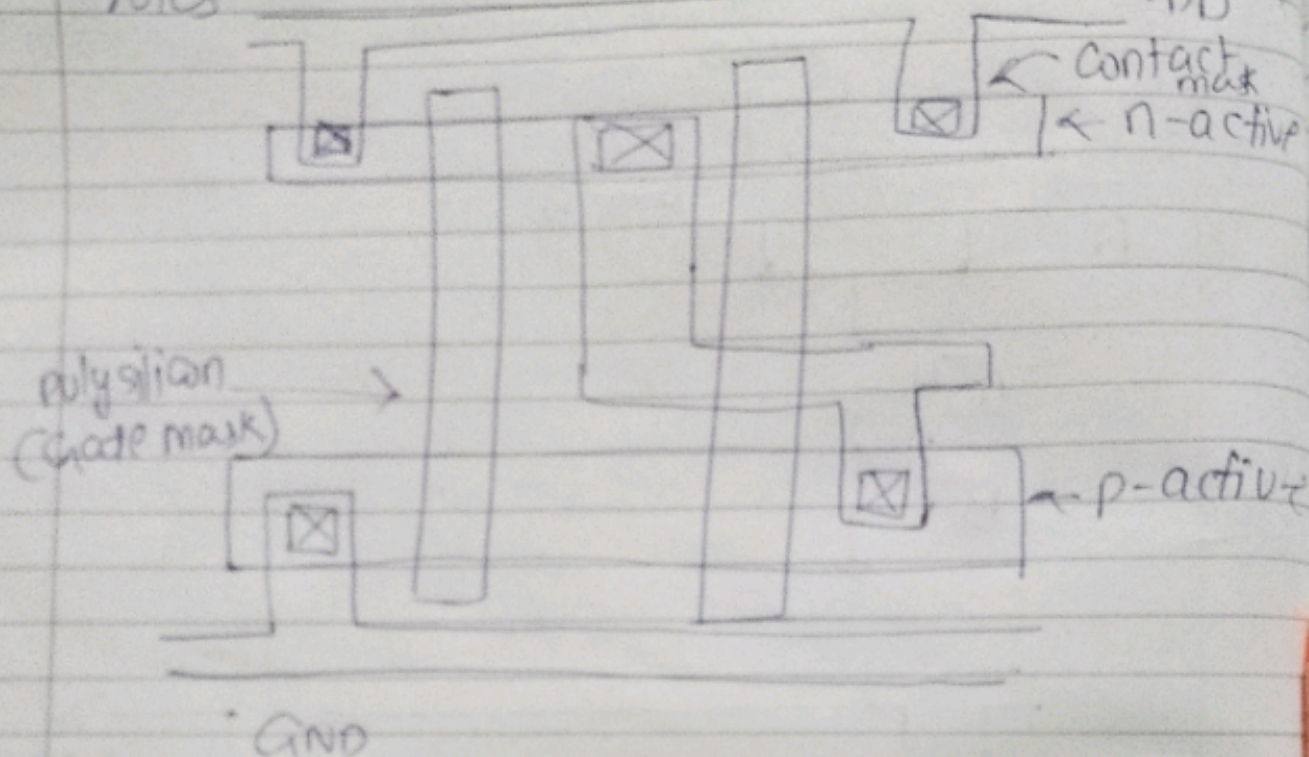
32. Draw two input NOR gate and its equivalent inverter if $(\frac{w}{L})$ ratio of all transistors in NAND is 20



Q. Draw Layout of NAND gate using lambda rules



34) Draw Layout of NOR gate using lambda rules



Q.35) Numerical based on finding of Noise Margin, Propagation delay etc. eg:-

Q.1) find noise margins for CMOS Inverter for following specification

$$V_{DD} = 3.3V, V_{TON} = 0.6V, V_{TOP} = -0.7V$$

$$k_n = \mu_n C_{ox} \frac{W}{L} = 200 \mu A/V^2, k_p = 80 \mu A/V^2$$

$$N_{ML} = |V_{OL} - V_{IL}|$$

$$N_{MH} = |V_{OH} - V_{IH}|$$

$$V_{OL} = 0$$

$$V_{OH} = 3.3V$$

$$V_{IL} = \frac{k_n V_{TON} + V_{TOP} \pm V_{DD} + 2V_o}{1 + k_R}$$

$$k_R = \frac{k_n}{k_p} = \frac{200}{80} = 2.5$$

$$V_{o1} = 2.5 \times 0.6 + (-0.7) + 3.3 + 2V_o$$

$$= -0.714 + 0.571V_o$$

$$= 0.571V_o - 0.714$$

$$I_{D \text{ sat}} = I_{D \text{ splin}}$$

$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS1} - V_{thn})^2 = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(V_{GS2} - V_{thp}) V_{DS2} - \frac{V_{DS2}^2}{2} \right]$$

$$\frac{1}{2} k_n [(V_i - 0.6)^2] = k_p [(V_i - V_{DD} - V_{thp})(V_o - V_{DD}) - \frac{(V_o - V_{DD})^2}{2}]$$

$$V_i = V_{IL}$$

$$\frac{1}{2} k_n [V_{IL} - 1]^2 = k_p [(V_{IL} - 3.3 + 0.7)(V_o - 3.3) - \frac{(V_o - 3.3)^2}{2}]$$

$$V_{IL} = 0.571V_o - 0.71$$

$$\frac{2.5}{2} (0.571V_o - 0.71 - 0.6)^2 = \left[(0.571V_o - 0.71 - 3.3 + 0.7)(V_o - 3.3) - \frac{(V_o - 3.3)^2}{2} \right]$$

$$1.25 (0.571V_o - 1.31)^2 = \left[(0.571V_o - 3.31)(V_o - 3.3) - \frac{V_o^2 - 2 \times 3.3V_o + 3.3^2}{2} \right]$$

$$= 1.25 [0.571V_o^2 - 2 \times 0.571V_o \times 1.31 + 1.31^2] =$$

$$\left[0.571V_o^2 - 3.31V_o - 0.571V_o \times 3.3 + 3.3 \times 3.31 - \frac{V_o^2 - 6.6V_o + 2 \times 3.3^2}{2} \right]$$

$$V_o = 3.11V$$

$$V_{IL} = 0.57 \times 3.11 - 0.71$$

$$V_{IL} = 1.0V$$

$$V_{IH} = \frac{V_{TOP} + V_{DD} + K_R (2V_0 + V_{ION})}{H + K_R}$$

$$= \frac{-0.7 + 3.3 + 25(2V_0 + 0.6)}{3.5}$$

$$V_{IH} = 1.43V_0 + 1.17$$

$$I_{n,lin} = I_{n,sat}$$

$$K_n \left[(V_{IH} - 0.6)V_0 - \frac{V_0^2}{2} \right] = \frac{K_p}{2} \left[(V_{IH} - V_{DD} - V_{TOP})^2 \right]$$

$$5 \left[(1.43V_0 + 1.17 - 0.6)V_0 - \frac{V_0^2}{2} \right] = (1.43V_0 - 3.3 + 0.7)^2$$

$$V_0 = 0.27$$

$$V_{IH} = 1.43 \times 0.27 + 1.17$$

$$V_{IH} = 1.556V$$

$$NML = [V_{OL} - V_{IL}] = 1.0 - 1.06$$

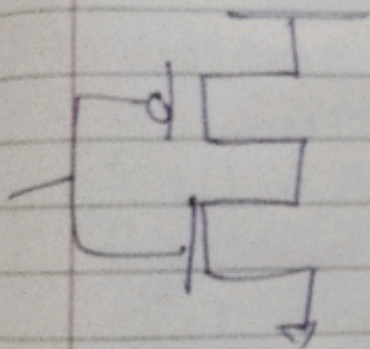
$$= 1.06$$

86) Implement give logic function using standard CMOS, PTL, Transmission gate pseudo NMOS

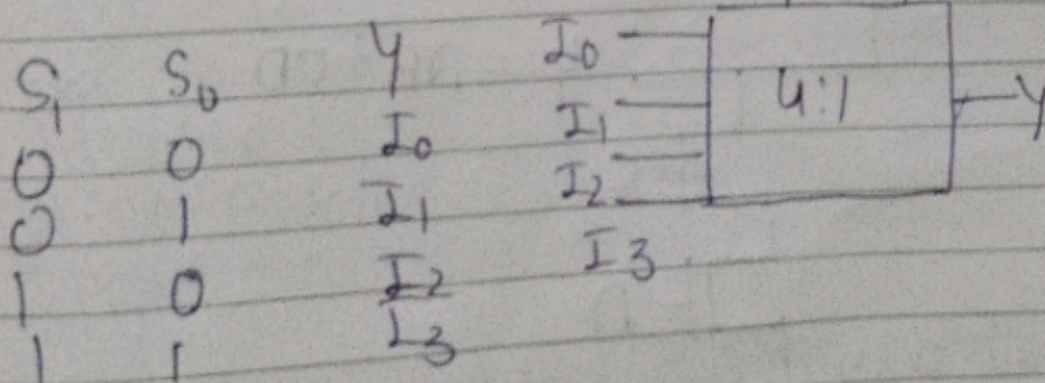
Standard CMOS

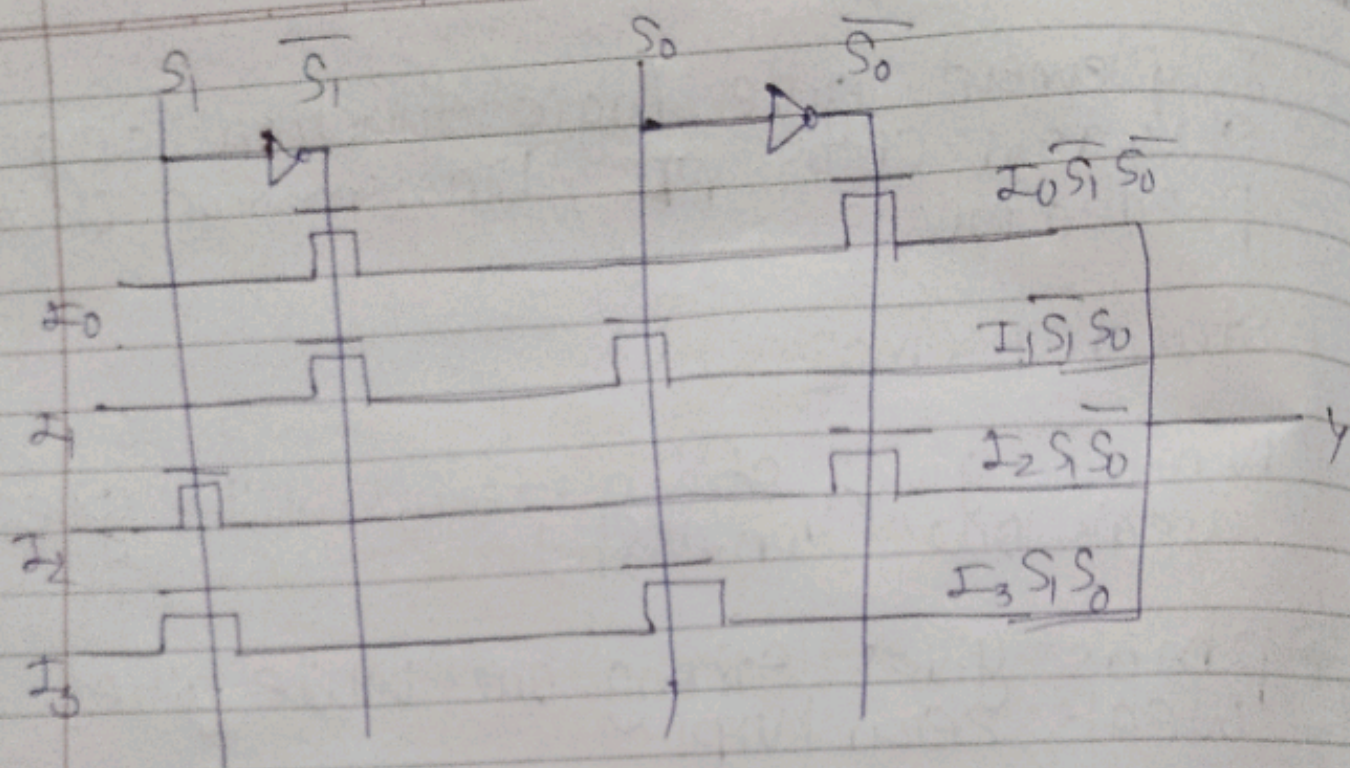
NMOS gives strong 'zero' while gives weak 'one' ($V_{DD} - V_{thn}$)

PMOS gives strong 'one' while gives weak zero $|V_{thp}|$



* Implement 4:1 mux using PTL

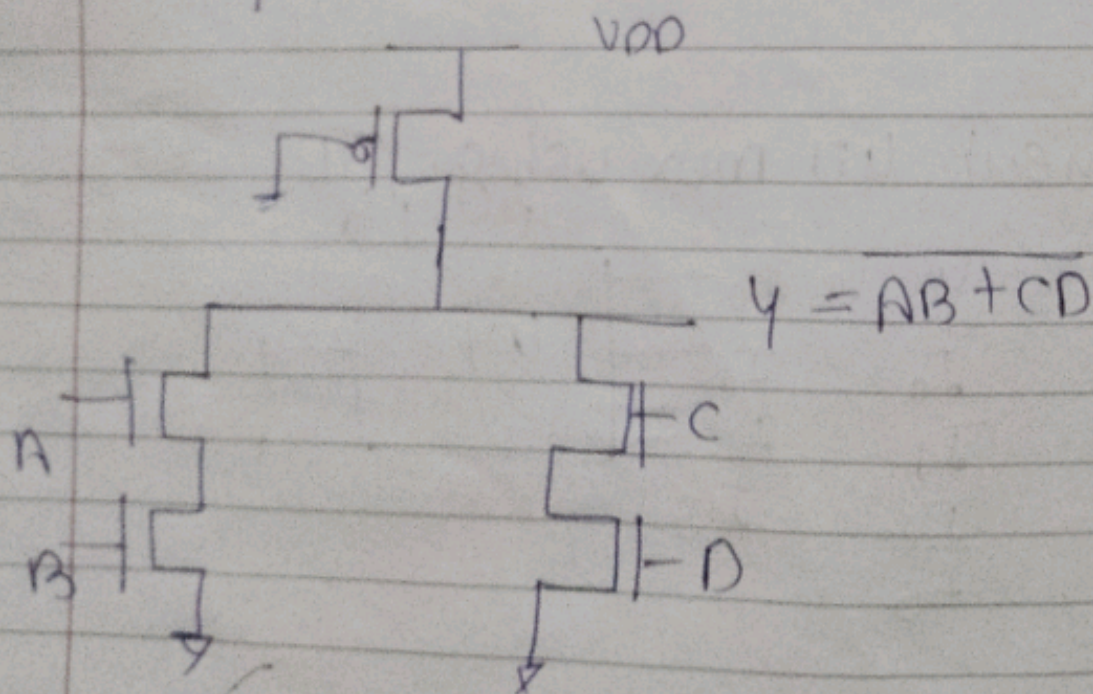




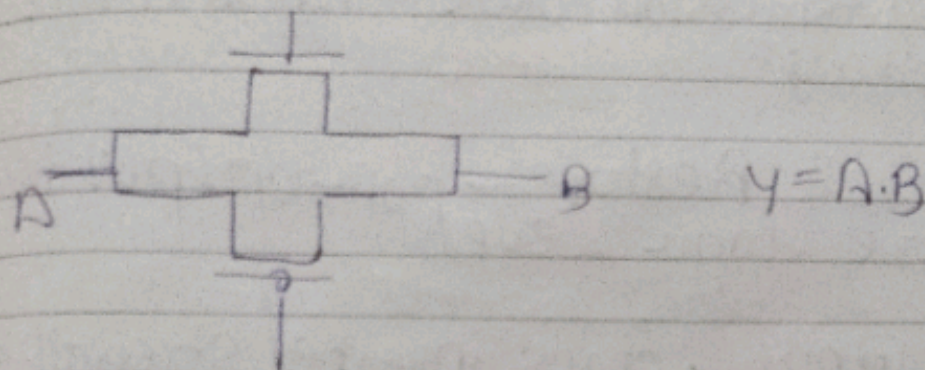
* Dynamic Logic

pseudo NMOS Logic

$$y = \overline{AB + CD}$$



* Transmission Gate logic



Q31) what are limitations of pass Transistor logic and how to overcome the same
Ans Limitations of pass Transistor logic (PTL)

- 1) Voltage Drop / Threshold loss
 - PTL often suffers from reduced output voltage levels because a pass transistor cannot pass a full logic '1' (V_{DD}) due to the threshold voltage V_{th} drop
- 2) weak Driving Capability
 - pass transistors have lower current driving ability compared to standard CMOS inverters
- 3) Noise Margin Reduction
 - More sensitive to noise
- 4) Delay increases - cascading transistors slows circuit

ways to overcome

- 1) Use Transmission Gate - NMOS + PMOS pass Full Voltage
- 2) Add level Restorer - pull-up / pull-down to restore logic level
- 3) Buffer output - CMOS inverter strengthens signal
- 4) Limit Cascading - Avoid too many series pass transistors

Q.38) Draw layout of Complex Circuit

39) Explain working of MOSFET in linear region, saturation region, Cut-off region using output characteristics. Also, derive expression of Drain Current in linear region and Saturation region

i) Cut-off region

$$V_{GS} = 0 \quad V_{DS} = 0 \quad V_{BS} = 0 \quad V_{GS} < V_{TO}$$

ii) linear region

$$V_{GS} > V_{TO} \quad V_{DS} < V_{GS} - V_{TO}$$

iii) Saturation Region

$$V_{GS} > V_{TO}$$

$$V_{DS} \geq V_{GS} - V_{TO}$$

$$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TO})^2$$

linear region

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TO}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Linear region

charge per unit area in channel

$$Q_n = -C_{ox}(V_{GS} - V_T)$$

$$dQ = -C_{ox}(V_{GS} - V(x) - V_T)W dx$$

Current due to drift velocity

$$I_D = Q_n v$$

Velocity

$$v = \mu_n E$$

Electric field

$$E = \frac{dv}{dx}$$

$$I_D = W \mu_n C_{ox} (V_{GS} - V(x) - V_T) \frac{dv}{dx}$$

$$I_D dx = W \mu_n C_{ox} (V_{GS} - V(x) - V_T) \cdot dv$$

Source $V = 0$

Drain $V = V_{DS}$

$$\int_0^L I_D \cdot dx = W \mu_n C_{ox} \int_0^{V_{DS}} (V_{GS} - V - V_T) \cdot dv$$

$$I_{DL} = W \mu_n C_{ox} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Saturation

$$V_{DS} = V_{GS} - V_T$$

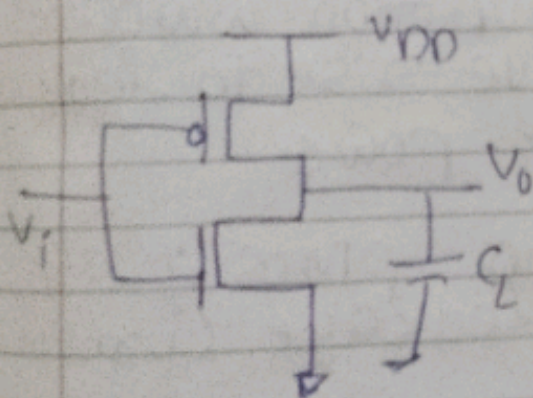
$$I_D = K_n [(V_{GS} - V_T)(V_{GS} - V_T) - \frac{(V_{GS} - V_T)^2}{2}]$$

$$I_D = K_n \left[(V_{GS} - V_T)^2 - \frac{(V_{GS} - V_T)^2}{2} \right]$$

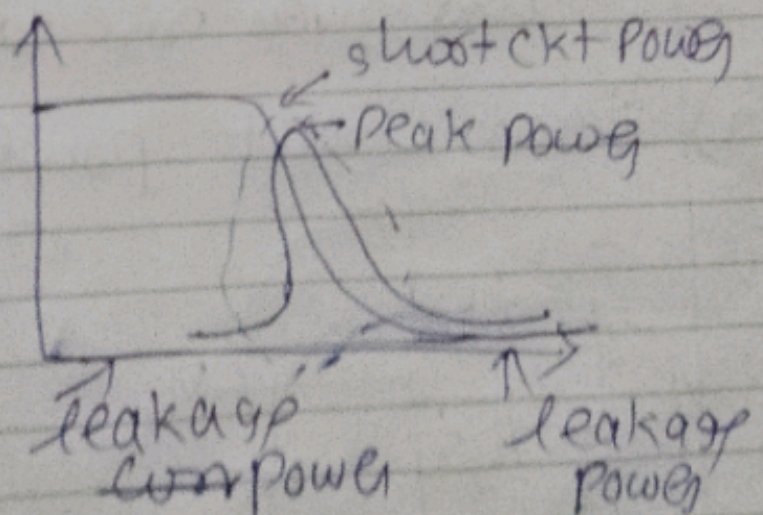
$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2$$

$$V_{DS} \geq V_{GS} - V_{T0}$$

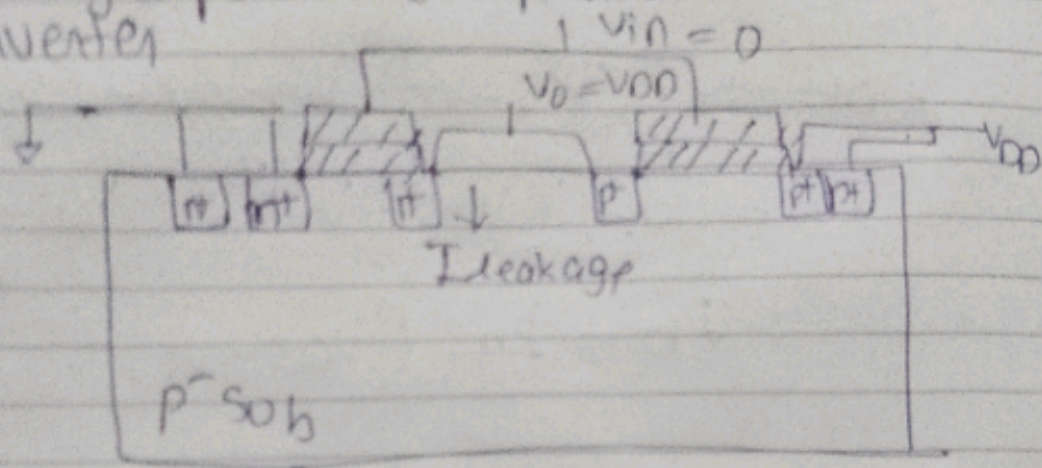
3.10) what are the Various Sources of power dissipation in CMOS Inverter? Explain each Source in detail



- i) Leakage power
- ii) Short Circuit power
- iii) Switching power (Dynamic power)



1) Leakage power Dissipation in CMOS Inverter



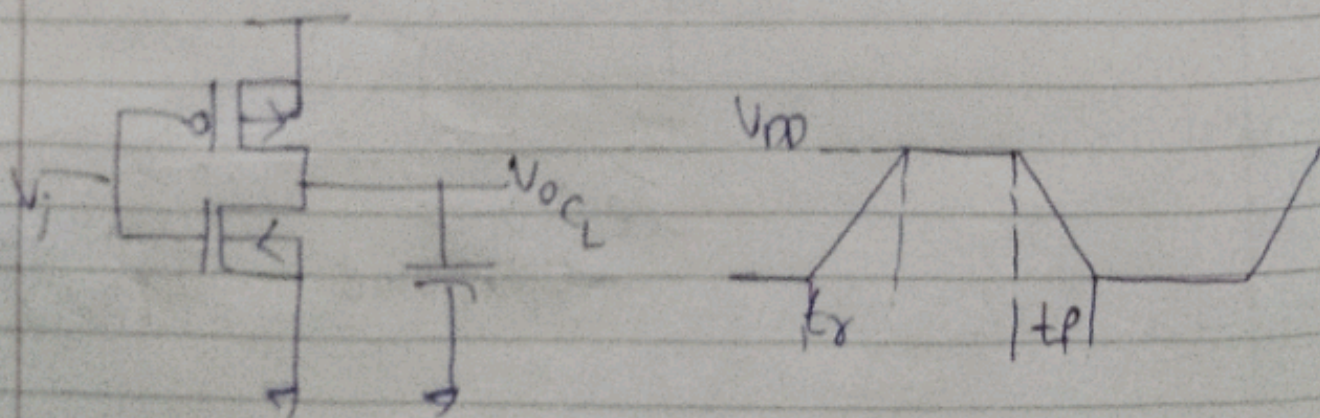
$$V_i = 0$$

$$V_o = V_{DD}$$

When $V_i = 0$, $V_o = V_{DD}$
In this case Drain of NMOS & Substrate of PMOS forms PN junction which is reversed bias This give rise to leakage Current and leakage power

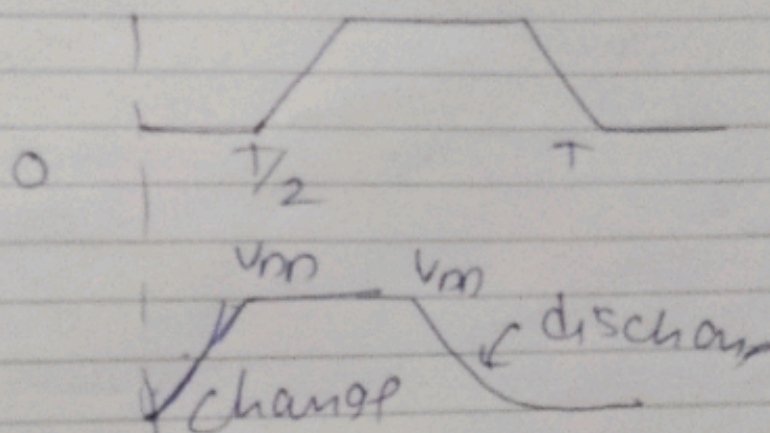
When $V_i = V_{DD}$, $V_o = 0$ PMOS is off and given rise to leakage Current

2) Short Circuit power Dissipation



when input is having finite t_r and t_f due to this both transistor are on depending duration of t_r and t_f . When both transistor are short circuit flows and give rise to Short Circuit power dissipation this power is proportional

iii) Switching power dissipation (Dynamic power Dissipation)



$$P_{avg} = \frac{1}{T}$$

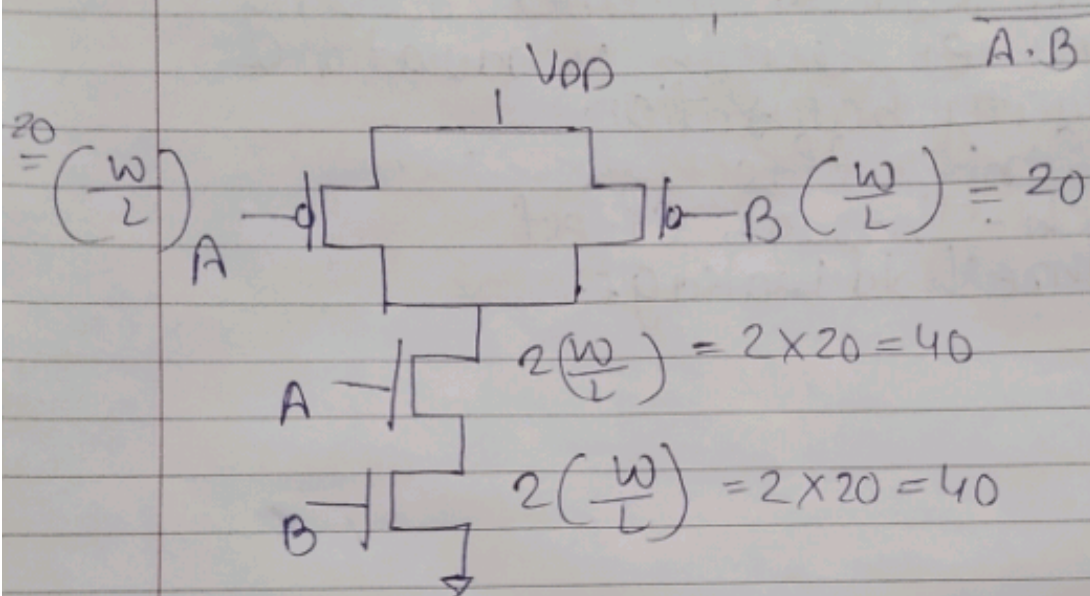
$$= \frac{1}{T} \left[\int_0^{T/2} V_{out} C_L \frac{dV_o}{dt} dt + \int_{T/2}^T V_{out} C_L \frac{dV_o}{dt} dt \right]$$

$$= \frac{1}{T} \left[C_L \frac{V_{out}^2}{2} + C_L \frac{V_{out}^2}{2} \right]$$

$$= \frac{1}{T} C_L V_{DD}^2$$

$$= C_L f_{CLK} V_{DD}^2$$

30. Draw two input NAND gate and its equivalent inverter if $(\frac{w}{L})$ ratio of all transistors in NAND gate is 20



32. Draw two input NOR gate and its equivalent inverter if $(\frac{w}{L})$ ratio of all transistors in NAND is 20

